Oecologia (1990) 84:93–98

Oecologia

Autumnal leaf conductance and apparent photosynthesis by saplings and sprouts in a recently disturbed northern hardwood forest

Jeffrey S. Amthor*, David S. Gill, and F. Herbert Bormann

School of Forestry and Environmental Studies, Yale University, New Haven, CT 06511, USA

Received September 22, 1989 / Accepted in revised form March 14, 1990

Summary. Leaf surface conductance and apparent photosynthesis were measured during late summer and autumn on saplings and sprouts of pin cherry (*Prunus pensylvanica*), yellow birch (*Betula alleghaniensis*), American beech (*Fagus grandifolia*), and sugar maple (*Acer saccharum*) naturally revegetating a site in the northern hardwood forest 5 years following a commercial whole-tree harvest. Prior to the disturbance (i.e., the harvest) the site was codominated by American beech, sugar maple, and yellow birch, whereas after the disturbance pin cherry was the dominant species. Conductance and photosynthetic rate of pin cherry leaves were comparatively high while those of American beech and sugar maple were low. Pin cherry retained green, physiologically active leaves longer into autumn than American beech and sugar maple. The rates and seasonal duration of leaf gas exchange on the disturbed site were therefore greater than they would have been had the site not become dominated by pin cherry.

Key words: Forest disturbance – Hubbard Brook – Northern hardwood ecosystem – Photosynthesis – Stomatal conductance

A dramatic shift in species composition occurs during the first several years following clear cutting in the mature northern hardwood forest of central New Hampshire, USA (Marks 1974). The recently disturbed site is often dominated by saplings of rapidly growing exploitive species[1] such as pin cherry and raspberry, but the conservative species American beech and sugar maple and the intermediate species yellow birch, which codominated the precut forest, are also present as saplings and sprouts (Bormann and Likens 1979). Accompanying the changes in species composition following a disturbance are changes in leaf phenology. Our preliminary observations suggest that leaves of exploitive species on a disturbed site remain green longer than leaves of conservative species occurring in the canopy of the adjacent intact forest and as saplings and sprouts in the cut-over site.

There have been few comparisons of leaf physiology of northern hardwood forest species *in situ* (Jurik et al. 1988), and none that we know of compared exploitive and conservative tree species on a recently disturbed site. Two specific research questions were addressed in this field study. (1) Are the *in situ* rates of gas exchange of leaves of exploitive species greater than conservative species, and if so, by what magnitude? (2) Are the leaves of exploitive species physiologically active later into autumn than leaves of conservative species as is indicated by the differences in leaf phenology? To answer these questions we measured leaf surface conductance and apparent photosynthesis during late summer and autumn on green and senescing (i.e., nongreen) leaves of four species in a naturally regenerating 5-year-old stand dominated by pin cherry.

Study site

This study was conducted on Watershed 5 at the Hubbard Brook Experimental Forest (HBEF) in the White Mountains of central New Hampshire, USA during late summer and autumn of 1988. The HBEF (43°56′ N, 71°45′ W) has been described in detail (Bormann and Likens 1979). Watershed 5 (22 ha) has a slope of about 12–13° and a SSE aspect. The bole and crown of all trees larger than 10 cm diameter were removed from the watershed in a commercial "whole-tree" harvest during autumn 1983 through summer 1984 with our study area being cut in 1983. Our leaf gas exchange measurements

* *Present address and address for offprint requests:* US Water Conservation Lab, 4331 E. Broadway, Phoenix, AZ 85040, USA

[1] The exploitive-intermediate-conservative classification scheme is based on functional similarities (i.e., growth rates, resource utilization patterns, age at first reproduction, etc.) of species within each group. It is roughly equivalent, but preferable, to the "early-mid-late successional" classification based on apparent similarities in temporal co-occurrence. This temporal classification is often misleading because most "late successional" species are present during the first year of forest ecosystem development following disturbance (Bormann and Likens 1979)

Table 1. Species composition and relative basal area on a 625 m^2 plot ("leaf phenology plot") immediately adjacent to the leaf gas exchange plot during November 1988 (5 years after the whole-tree harvest). The basal area of all living woody stems was measured at 10 cm aboveground in 90 plots (each 1 × 1 m). The total basal area was 0.0154 m^2 m^{-2}. Species accounting for at least 1% of total area are arranged alphabetically within functional groups. Botanical nomenclature follows Fernald (1950) except for *Betula alleghaniensis* Britt. (= *B. lutea* Michx.)

Functional group	Scientific name	Common name	Relative (%) basal area
Exploitive:	*Prunus pensylvanica* L.	Pin cherry	56.9
	Rubus allegheniensis Porter	Blackberry	1.1
	Rubus idaeus L.	Raspberry	2.7
Intermediate:	*Acer pensylvanicum* L.	Striped maple	7.3
	Betula alleghaniensis Britt.	Yellow birch	5.4
	Betula papyrifera Marsh.	Paper birch	1.3
	Fraxinus americana L.	White ash	3.4
	Populus tremuloides Michx.	Quaking aspen	1.8
Conservative:	*Acer saccharum* Marsh.	Sugar maple	14.2
	Fagus grandifolia Ehrh.	American beech	3.0

were confined to sprouts and saplings on a 625 m^2 plot ("gas exchange plot") at an elevation of about 520 m that remained undisturbed following the cut. The watershed and adjacent forest had not been previously cut since about 1917 (Bormann et al. 1970).

Species composition at the site 5 years after cutting was typical of recently clear-cut northern hardwood stands (Marks 1974; Safford and Filip 1974; Bormann and Likens 1979; Messier and Bellefleur 1988) with pin cherry as the dominant, ranging up to 5 m in height, and with species of *Rubus, Betula, Fagus, Acer, Fraxinus,* and *Populus* also present (Table 1). The forest surrounding the watershed, and the watershed prior to the cut, was codominated by American beech, sugar maple, and yellow birch, with some white ash present also.

Daily minimum and maximum air temperatures were recorded at a meteorological station at about the same elevation and aspect of the study site, but 0.8 km to the east. Rain amount was monitored at this station and at a station adjacent to (west of) the study site. Rain was well distributed during late summer and autumn 1988, and probably sufficient to maintain a good plant water status during the study.

Methods

Leaf surface conductance of water vapor diffusion and apparent photosynthesis were measured for arbitrarily selected leaves of pin cherry, American beech, yellow birch, and sugar maple. Only one leaf on a tree was sampled. No attempt was made to return to the same trees (or leaves) on subsequent measurement dates.

For our purposes leaves were assigned to one of two classes: green and nongreen. The classification was based on visual observation at the time of the gas exchange measurement. Leaves that had not appeared to have experienced much net chlorophyll loss during autumn were considered green. Leaves containing little or no visible green pigments were considered nongreen. Most of the measurements were made on green leaves because it was our goal to determine whether the green color of a leaf could be used as an indicator of physiological activity during autumn.

Data reported here for green leaves (n = 314) were collected on 25 August (day of the year [DOY] 238), 6–8 September (DOY 250–252), 14–16 September (DOY 258–260), 21–22 September (DOY 265–266), 27–28 September (DOY 271–272), 4 and 6 October (DOY 278 and 280), 12–13 October (DOY 286–287), 16 October (DOY 290), and 19 October (DOY 293). Data within each of the groups identified were pooled.

Measurements were made with a LI-COR LI-6200 portable photosynthesis system between 1200 and 1700 h EST at ambient CO_2 concentration, temperature, and humidity. The mid-section (usually 1300–1500 mm^2) of a leaf was enclosed in a 1-l cuvette of the LI-6200. After the CO_2 concentration started to decline following closure of the cuvette three consecutive 10-s measurements were made. The second of the three measurements was used in this study to estimate photosynthesis and conductance. The first and third measurements were used to determine if the leaf was in a reasonably steady-state condition during the measurement (if not, the observation was discarded) although a steady state may be more of an exception than the rule for leaves *in situ*. The LI-6200 IRGA was calibrated on the site with a span gas of known CO_2 concentration. Leaf boundary layer conductance of water vapor when inside the cuvette was about 60 mm s^{-1}.

The PPFD was measured concomitantly with gas exchange by a quantum sensor attached to the leaf cuvette in the plane of the leaf in order to determine conductance and photosynthesis in response to PPFD *in situ*. Care was taken to ensure that the sensor and leaf were experiencing similar PPFD's during measurements. Leaf orientation was not altered during gas exchange measurements. PPFD was "varied" by measuring leaves at different locations in the canopy and having different angles with respect to direct beam solar radiation as well as under different sky conditions. Low-light measurements were usually obtained with leaves facing North or North-East while high-light measurements were made on South to South-West facing leaves. Data reported here were not collected soon after large changes in incident PPFD due to moving sunflecks or rapid changes in sky conditions.

Results and discussion

Comparative autumnal conductance

In this field study conductance of green leaves was not influenced by light above PPFD's of about 0.1 mmol

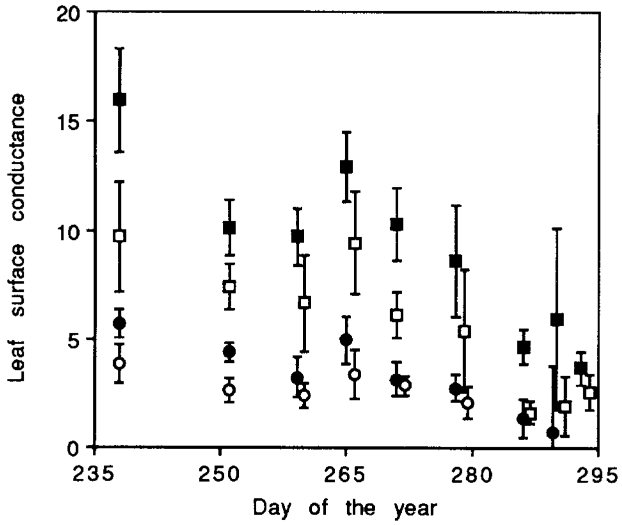

Fig. 1. Mean surface conductance of water vapor diffusion (mm s^{-1}) by green leaves during late summer and autumn. Only measurements made when the PPFD was at least 0.1 mmol m^{-2} s^{-1} are included in the means. The bars show 95% confidence intervals about the means. Dates for which no data are shown for a species indicate that no green leaves of that species were observed on the gas exchange plot on that date. Some points have been moved horizontally to avoid overlap of the confidence interval bars. Symbols: ■ pin cherry; □ yellow birch; ● American beech; ○ sugar maple

m^{-2} s^{-1} (data not shown). That is, except in heavy shade, stomata were not appreciably affected by light level. This independence of conductance from PPFD (above 0.1 mmol m^{-2} s^{-1}) is consistent with previous reports for hardwood trees in the field (Federer and Gee 1976; Turner and Heichel 1977) and was the case for all four species during all the measurement periods.

There were large differences in conductance by green leaves among the four species examined when PPFD was at least 0.1 mmol m^{-2} s^{-1} (Fig. 1). Pin cherry green leaf conductance was consistently about 4 times that of sugar maple, about 3 times that of American beech, and about 1.5 times that of yellow birch from DOY 238 to DOY 280. Measured conductance of green pin cherry leaves during this time, which was often in excess of 10 mm s^{-1}, was greater than previously reported values for conservative hardwood species (Hinckley et al. 1978; Körner et al. 1979), but similar to, or slightly greater than, our measurements made on pin cherry saplings at a nearby site during late summer and autumn 1987 (unpublished).

When the leaves were senescing (losing green color) conductance was greatly reduced, presumably due to the loss of stomatal function (Reich 1984). Conductance of leaves of all four species that had senesced was 30%, or less, of that of fully green leaves of the same species at the same time (data not shown). Generally, conductance of reddish pin cherry leaves exceeded that of yellowing American beech and sugar maple leaves, reflecting the species differences in conductance by green leaves. A sharp drop in leaf conductance of northern hardwoods during senescence (net loss of chlorophyll) has been previously noted (Gee and Federer 1972).

Seasonal trends in conductance

As a percentage of late summer (DOY 238) values conductance of the four species paralleled each other through DOY 280 (Fig. 1). The week-to-week variability in intraspecific conductance during that time may have been related to daily and weekly changes in environmental conditions.

By DOY 286 all sugar maple leaves on the gas exchange plot had senesced, and many had abscised (see also Fig. 2). At this time conductance of green leaves on pin cherry, yellow birch, and American beech were about 25% the values on DOY 238. Air temperatures of 0° C or less were recorded on DOY 280, 281, 287, and 288, which may have contributed to the changes in conductance after DOY 280 (Fig. 1).

By DOY 293 all American beech leaves on the gas exchange plot had senesced and conductance of green pin cherry leaves had fallen to 3–4 mm s^{-1}. Note, however, that these values of pin cherry green leaf conductance were similar to, or higher than, values for sugar maple during the entire study period. Conductance of green leaves of yellow birch was 2–3 mm s^{-1} on DOY 293.

Only a few yellow birch leaves retaining green color were found on the gas exchange plot by DOY 294 and

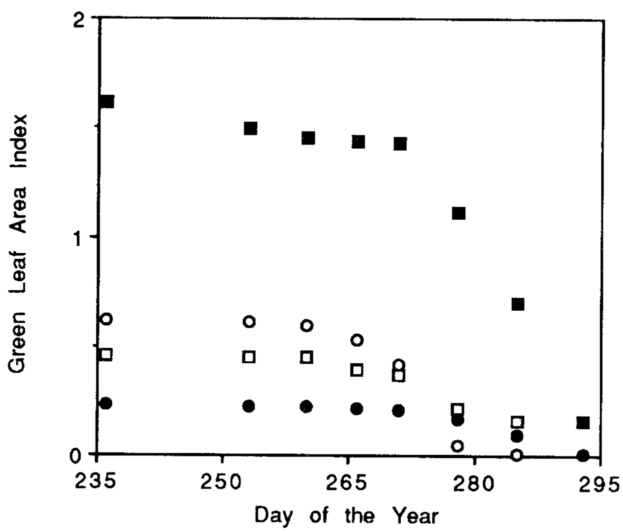

Fig. 2. Mean green LAI (m^2 green leaves m^{-2} ground area) during late summer and autumn on the leaf phenology plot (Table 1). The number of green leaves on three marked shoots on each of nine randomly selected individuals of each species were periodically counted. A green leaf index (GLI) was calculated by dividing the number of mature green leaves at each date by the total number of leaves produced over the course of the season; the GLI expressed the proportion of seasonal total LAI present as mature green leaves at a given time. Seasonal total LAI was estimated during the first week of November, after all leaves had abscised, by counting the number of leaf intercepts of each species along thin probes inserted randomly into the forest floor at 225 locations throughout the plot. This technique is essentially the point quadrat method initially described by Warren Wilson (1959). Values of green LAI for each species at a given time were calculated as the product of the GLI at that time and the seasonal total LAI for that species. When no value is shown for a species on a given date the green LAI was zero for that species. Symbols see Fig. 1

they had gas exchange rates less than one third the rates measured during the previous day. Green pin cherry leaves were rare at that time (see also Fig. 2) although the larger of the pin cherry trees retained green to reddish-green leaves at the top of the main stem beyond that date, but gas exchange measurements were not made on those leaves because of their inaccessibility.

Comparative autumnal photosynthesis

An important finding of this field study is the higher apparent light saturation points of pin cherry leaves compared to the other three species (e.g., Fig. 3). During most measurement periods American beech and sugar maple apparent photosynthesis was saturated at a PPFD of no more than 1.0 mmol m^{-2} s^{-1} while pin cherry photosynthesis increased with increasing PPFD well above 1.0 mmol m^{-2} s^{-1} during all but the mid-October period (after DOY 280). Yellow birch was intermediate among the species with respect to apparent light-saturation point.

Typical high-light (PPFD > 1.0 mmol m^{-2} s^{-1}) leaf photosynthetic rates during late summer were: 15–19, 10–14, 6–9, and 5–7 μmol CO_2 m^{-2} s^{-1} for pin cherry, yellow birch, American beech, and sugar maple, respectively. These values of pin cherry apparent photosynthesis under high light exceed rates reported for most hardwoods, but are below values for some *Populus* and *Malus* species under intensive culture (Nelson 1984).

Photosynthesis at a PPFD of 1.0 mmol m^{-2} s^{-1} was estimated from the logarithmic functions (see Fig. 3). These values were greatest for pin cherry, followed by yellow birch, which was followed by American beech and sugar maple throughout the study period (Fig. 4). Even at lower PPFD's (<0.5 mmol m^{-2} s^{-1}) pin cherry often had higher rates of leaf photosynthesis than the other three species (e.g., Fig. 3), indicating that during most of the day pin cherry assimilated more CO_2 per unit leaf area. In addition, pin cherry dominated the top of the canopy, shading the other species.

Senescing (nongreen) pin cherry leaves typically had apparent photosynthetic rates 30%, or less, those of green pin cherry leaves for a given light level (data not shown). Senescing leaves of yellow birch, American beech, and sugar maple had even lower rates of photosynthesis compared to green leaves of the same species at the same time. In fact, negative CO_2 exchange rates were often observed, even under high PPFD, i.e., respiratory CO_2 efflux exceeded photosynthetic CO_2 assimilation. Rapid rates of respiration in senescing leaves may be related to metabolism associated with the breakdown and extraction of nitrogenous compounds.

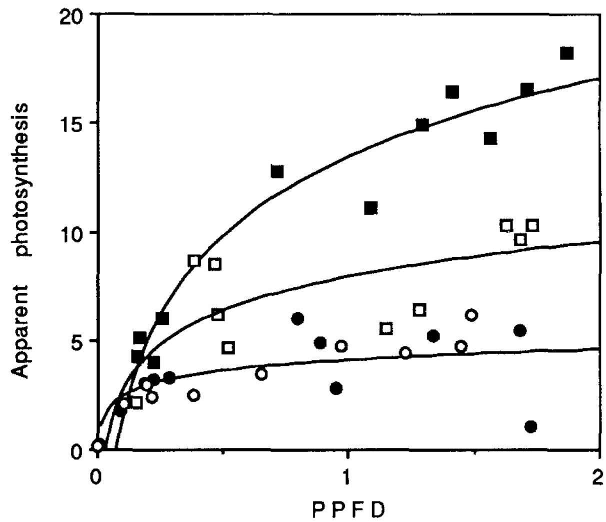

Fig. 3. Apparent photosynthesis (AP, μmol CO_2 m^{-2} s^{-1}) by green leaves as a function of PPFD (mmol m^{-2} s^{-1}) on DOY 271–272. These data are typical of the late summer and early autumn measurements. The solid lines show logarithmic functions used to summarize the data for pin cherry, yellow birch, and American beech (from top to bottom, respectively). They are drawn here as examples. The equations of the lines drawn are: $AP = 13.42 + 12.08 \log_{10}$ (PPFD), $AP = 7.92 + 5.23$ $\log_{10}$(PPFD), and $AP = 4.07 + 1.61$ $\log_{10}$(PPFD), for pin cherry, yellow birch, and American beech, respectively. The equation for sugar maple during this sampling period is $AP = 4.34 + 1.90 \log_{10}$(PPFD), but that line is not drawn because it nearly overlays the line for American beech. Logarithmic functions were determined for each species for each sampling period and were used to estimate photosynthesis (with a 95% confidence interval) at a PPFD of 1.0 mmol m^{-2} s^{-1} (Fig. 4). Logarithmic functions are not applicable to very low PPFD conditions. Symbols see Fig. 1

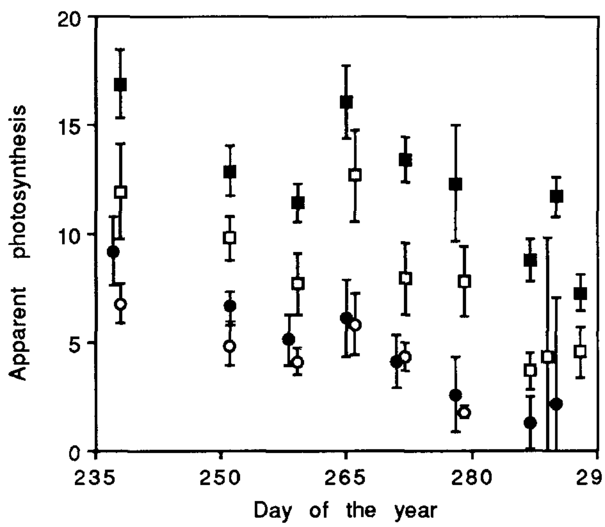

Fig. 4. Apparent photosynthesis (μmol CO_2 m^{-2} s^{-1}) by green leaves at a PPFD of 1.0 mmol m^{-2} s^{-1} (see Fig. 3) during late summer and autumn. Other details are as given with Fig. 1. Symbols see Fig. 1

Seasonal trends in photosynthesis

From DOY 250 through DOY 272 the ratios of photosynthetic rate (at a PPFD of 1.0 mmol m^{-2} s^{-1}) among the species were similar to the ratios on DOY 238 (Fig. 4). Thereafter, pin cherry photosynthesis on a leaf area basis was always greater than American beech and sugar maple, and usually greater than yellow birch, as a percentage of DOY 238 values. On DOY 278 and 280, for example, pin cherry photosynthesis was 72%

that of DOY 238 values whereas American beech and sugar maple rates had declined to only 27–28% of the DOY 238 values. By DOY 286–287 pin cherry photosynthesis was 52% the rates measured on DOY 238 (and during summer) while all sugar maple leaves on the gas exchange plot had senesced and American beech photosynthesis had declined to 14% of DOY 238 values.

On DOY 293, when green leaves of only pin cherry and a few yellow birch existed on the gas exchange plot (see also Fig. 2), photosynthesis by green leaves of pin cherry was 43% as great as values on DOY 238 and was comparable to, or exceeded, that of American beech and sugar maple measured during the previous 6 weeks (Fig. 4). Compared to earlier rates by American beech and sugar maple, yellow birch green leaf photosynthesis also was still high at this time, but lower than that of pin cherry, as it was during the entire study period.

Thus, the relative differences in leaf photosynthetic rates among species noted in the DOY 238 data were exaggerated during autumn when photosynthesis by the conservative species declined earlier than photosynthesis by pin cherry (Fig. 4).

Leaf senescence

While the data presented in Figs. 1, 3, and 4 are for "green" leaves it is likely that senescence was occurring in some of those leaves, especially towards the end of the study period. The early stages of leaf senescence are not easily discerned by visual observations of color and some loss of stomatal control and photosynthetic capacity may occur in green leaves during autumn. Such early stages of senescence may have contributed to an increase in variability and an overall decline in conductance and apparent photosynthesis for all four species during autumn (Reich 1984). Nonetheless, leaves could be objectively classified as green or nongreen and we always observed marked differences in conductance and photosynthesis between green and nongreen leaves of a given species at a given time. Moreover, with few exceptions, our measurements of green leaves were consistent (i.e., variability was small) during a given sampling period and differences among species were clear. Finally, as leaves were senescing during autumn our measurements of green leaf physiology may have been from different phenotypes within the populations sampled. Our results therefore may have been affected by phenotypic differences expressed during autumn, if those differences indeed existed and were expressed.

Seasonal trends in green leaf display

To put our individual leaf measurements in context it is useful to consider the trends in autumnal leaf display among the four species studied. That is, while green leaves of pin cherry have greater rates of gas exchange than the other species studied, this fact must be coupled with the green LAI of pin cherry at any particular time to appreciate the potential significance to the ecosystem of these greater leaf gas exchange rates.

Pin cherry had a greater green LAI than the other species on the leaf phenology plot throughout late summer and autumn (Fig. 2). For instance, green LAI on DOY 236 was 1.61, 0.47, 0.23, and 0.62, for pin cherry, yellow birch, American beech, and sugar maple, respectively. Leaves changed color slowly from DOY 236 to 271 for all species, but the green LAI declined rapidly after that, especially for sugar maple and pin cherry. By DOY 285 green LAI on the leaf phenology plot had decreased to 0.70, 0.16, 0.10, and 0.01 for pin cherry, yellow birch, American beech, and sugar maple, respectively. Note, however, that this value for pin cherry exceeded that of the other species measured during late summer. By DOY 293 pin cherry green LAI was still 0.16 whereas virtually all leaves of the other species had changed color and/or abscised.

Shoot development in pin cherry, and other exploitive species in the northern hardwood forest, is strongly indeterminate (Marks 1975; Bormann and Likens 1979). Leaf production in pin cherry and yellow birch (also largely indeterminate) continued throughout August, while leaf production was largely completed in the determinate species American beech and sugar maple during June. The leaves retained by pin cherry and yellow birch in autumn were produced primarily during July and August rather than during May and June (data not shown; details of phenological observations will be presented in a later communication). Differences in environmental conditions at the time of leaf development may account, in part, for the observed differences among species in gas-exchange physiology during autumn.

Conclusions

The comparative measurements presented here document the differences in magnitude and seasonal duration of leaf-level gas exchange among conservative, intermediate, and exploitive species. To our knowledge these are the only such comparative data from a recently disturbed northern hardwood forest. The results fit the current paradigm concerning the rates of leaf physiology between conservative (late successional) and exploitive (early successional) species (reviewed by Bazzaz 1979). Differences in seasonal duration of leaf physiological activity between exploitive and conservative tree species growing in natural ecosystems during recovery from disturbance, however, have apparently not yet been reported, nor have the consequences for ecosystem recovery been considered.

In summary, in situ leaf conductance of water vapor diffusion (Fig. 1) and apparent photosynthesis (Figs. 3 and 4) were greatest for the exploitive species, pin cherry, intermediate for yellow birch, and lower for the conservative species, American beech and sugar maple on a recently disturbed site in the northern hardwood forest. In addition, pin cherry trees retained green (Fig. 2), physiologically-active (Figs. 1 and 4) leaves later into autumn compared to American beech and sugar maple. These results, that exploitive species (1) have greater leaf-level rates of physiological gas exchange and (2) remain physiologically active longer into autumn, could have important consequences for ecosystem recovery following disturbance.

Acknowledgements. This is a contribution to the Hubbard Brook Ecosystem Study and was supported by an Andrew W. Mellon Foundation grant to FHB and NSF grant BSR 8406633. The HBEF is operated and maintained by the Northeastern Forest Experiment Station, USDA, Broomall, PA, USA

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